

TAGDALEN, Norway

WMO: 012210

Lat: 63.056N

Lon: 9.072E

Elev: 1362

StdP: 13.99

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.5	(c) 9.6	(d) -3.3	(e) 4.9	(f) 10.1	(g) -0.1	(h) 5.7	(i) 12.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	75.6	59.7	71.5	57.7	67.9	55.7	61.2	73.3	58.9	69.2	57.0	66.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	70.1	65.0	54.1	65.4	62.8	52.2	60.9	61.6	27.8	74.0	26.2	69.3	25.0	66.0	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) 2.4	(f) 81.0	(g) 5.9	(h) 5.1	(i) -1.8	(j) 84.7	(k) -5.3	(l) 87.7	(m) -8.6	(n) 90.6	(o) -12.9	(p) 94.3		
(4) N/A	(5) N/A	(6) N/A	DB	2.4	81.0	5.9	5.1	-1.8	84.7	-5.3	87.7	-8.6	90.6	-12.9	94.3	(4)
			WB	1.4	64.4	5.5	3.1	-2.6	66.7	-5.8	68.6	-8.9	70.3	-13.0	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	25.9	27.4	30.4	37.5	44.9	50.2	56.3	55.0	49.0	40.0	32.0	27.8		
	DBStd	12.82	7.78	7.69	6.48	6.53	8.13	6.53	7.13	5.58	5.89	6.62	7.86	8.18		
	HDD50	4311	747	632	607	377	206	76	17	15	88	317	540	689		
	HDD65	9230	1212	1052	1072	824	624	445	288	314	479	776	990	1154		
	CDD50	579	0	0	0	4	47	82	211	169	60	6	0	0		
	CDD65	20	0	0	0	0	0	1	17	2	0	0	0	0		
	CDH74	190	0	0	0	0	5	17	150	18	0	0	0	0		
	CDH80	26	0	0	0	0	0	0	26	0	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	58.9	5.7	5.4	5.6	3.1	3.1	4.3	4.0	5.2	5.7	5.4	5.2	6.1		
(15) Precipitation	PrecMax	74.4	16.0	12.3	14.0	8.7	6.5	8.2	6.7	11.6	16.2	10.7	14.0	13.5		
	PrecMin	39.5	0.2	0.1	1.4	0.7	1.8	1.7	1.2	1.8	2.0	1.1	0.2	1.4		
	PrecStd	8.5	3.7	2.8	3.4	2.0	1.1	1.7	1.5	2.4	3.4	2.0	3.9	2.8		
	PrecStd	8.5	3.7	2.8	3.4	2.0	1.1	1.7	1.5	2.4	3.4	2.0	3.9	2.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	46.1	48.5	62.5	73.7	76.6	83.2	76.5	69.0	60.8	51.4	45.6		
		MCWB	35.3	37.7	38.2	49.3	56.6	57.4	65.5	60.6	56.1	49.9	44.3	38.4		
	2%	DB	40.2	41.9	44.4	56.6	69.0	70.2	78.7	71.7	64.9	56.9	47.2	42.8		
		MCWB	34.3	34.6	36.3	44.1	53.7	56.3	62.0	58.3	54.0	47.2	40.8	36.6		
	5%	DB	37.8	39.4	41.5	51.4	64.6	65.6	73.6	68.4	61.4	52.4	44.6	40.4		
		MCWB	34.1	33.2	35.1	41.5	51.5	54.0	59.0	56.3	52.2	44.4	39.5	35.1		
	10%	DB	35.8	37.2	39.4	47.4	59.4	61.9	69.0	64.9	58.3	48.8	41.8	38.1		
		MCWB	32.8	32.6	34.5	39.0	48.8	52.0	57.5	55.2	50.6	42.3	37.2	34.0		
	0.4%	WB	38.0	38.7	40.3	50.0	57.9	59.8	67.3	62.1	57.9	52.8	45.6	40.1		
		MCDB	40.6	43.0	43.9	61.7	71.7	70.8	82.6	72.9	66.0	58.9	50.4	43.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.4	36.4	38.4	44.9	54.7	57.1	62.8	59.7	55.4	47.2	42.3	37.9		
		MCDB	38.1	39.7	41.6	55.1	67.1	69.5	76.3	69.3	62.2	55.4	45.8	41.6		
	5%	WB	34.7	34.8	36.9	42.2	51.7	54.7	60.5	58.0	52.9	45.0	39.8	35.9		
		MCDB	36.9	37.9	40.4	50.1	63.0	65.0	72.8	66.3	59.5	51.2	43.7	39.4		
	10%	WB	33.0	33.2	34.8	39.9	49.1	52.3	57.9	56.0	51.0	42.8	37.7	34.0		
		MCDB	35.9	36.4	38.8	46.4	59.4	60.8	67.7	62.9	57.3	48.1	41.3	37.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	11.5	13.3	15.2	14.3	14.7	14.1	12.6	10.6	8.6	8.5		
		MCDBR	9.4	10.6	13.7	19.3	23.2	21.8	22.3	19.8	17.4	15.5	11.4	10.4		
		MCWBR	6.6	6.9	8.6	10.5	12.3	12.0	11.3	10.5	10.1	9.5	7.7	7.7		
		MCDBR	8.0	9.3	11.2	17.5	22.3	20.7	21.7	18.1	16.3	14.0	10.1	10.3		
	5% WB	MCWBR	6.4	6.7	8.0	10.3	12.5	12.3	11.5	10.3	10.2	9.2	7.6	8.2		
		MCWBR	6.4	6.7	8.0	10.3	12.5	12.3	11.5	10.3	10.2	9.2	7.6	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.280	0.312	0.340	0.342	0.343	0.352	0.340	0.318	0.295	0.232	0.364		
	taud		2.026	2.198	2.235	2.296	2.369	2.443	2.450	2.482	2.530	2.418	2.092	2.671		
	Ebn at Noon		129	209	248	267	277	279	272	266	251	207	123	75		
	Edh at Noon		10	19	28	33	33	31	30	27	21	16	9	6		
(44) All-Sky Solar Radiation	RadAvg		54	229	651	1216	1557	1595	1473	1138	692	307	82	28		
	RadStd		9	37	82	82	87	130	128	109	74	39	15	3		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page